Task 1:

Install FileZilla Server on your Windows machine, create a user account, and set up a shared folder for FTP access.

Steps Performed

1. I opened the web browser and visited the official FileZilla website: https://filezilla-project.org/
2. I downloaded and installed FileZilla Server on my Windows computer.
3. I opened FileZilla Server and connected to its administration interface.
4. I opened the Users section.
5. I created a new FTP user account and entered a username and password.
6. I opened the Shared folders section.
7. I selected a folder on my computer to share through FTP.
8. I assigned the required read and write permissions to the user.
9. I saved the configuration settings.
10. I checked that the FileZilla Server service was running.
11. I noted the server's local IP address using the ipconfig command.
12. I tested the FTP connection using an FTP client with the server IP address, username, and password.

Result: FileZilla Server was installed, a user account was created, and a shared folder was configured for FTP access.

Task 2:

Use FileZilla Client to connect to your FTP server using the credentials you created, and upload any image file from your computer to the shared folder.

Answer:

Steps Performed

1. I opened the web browser and visited the official FileZilla website: https://filezilla-project.org/
2. I downloaded and installed FileZilla Server on my Windows computer.
3. I opened FileZilla Server and connected to its administration interface.
4. I opened the Users section.
5. I created a new FTP user account and entered a username and password.
6. I opened the Shared folders section.
7. I selected a folder on my computer to share through FTP.
8. I assigned the required read and write permissions to the user.
9. I saved the configuration settings.
10. I checked that the FileZilla Server service was running.
11. I noted the server's local IP address using the ipconfig command.
12. I tested the FTP connection using an FTP client with the server IP address, username, and password.

Result: FileZilla Server was installed, a user account was created, and a shared folder was configured for FTP access.

Task 3:

Download the uploaded file from your FTP server to a different folder on your computer using FileZilla Client, and verify the file size matches the original.

Steps Performed

1. I opened FileZilla Client on my computer.
2. I connected to the FTP server using the server IP address, username, and password.
3. I located the uploaded image file in the Remote Site section.
4. I selected a different folder on my computer in the Local Site section.
5. I right-clicked the uploaded file and selected Download.
6. I waited until the file transfer was completed.
7. I opened the destination folder and checked that the downloaded file was present.
8. I right-clicked the original image file and selected Properties to check its size.
9. I checked the downloaded image file's properties and compared its size with the original file.
10. I verified that both files had the same size.

Result: The uploaded image file was downloaded to a different folder, and its file size was verified against the original.

Task 4:

Research and list 3 real-world scenarios where SNMP is used in network management for apps or services you use (such as ISPs, campus WiFi, or large websites).

Answer:

1. Internet Service Providers (ISPs)

Internet service providers use SNMP to monitor routers, switches, and network devices. It helps them check bandwidth usage, network traffic, device status, and connection problems.

Example: When I use Wi-Fi at home to access YouTube or Instagram, my ISP can use SNMP to monitor supported network equipment and identify performance issues.

2. College Campus Wi-Fi

Colleges use SNMP to monitor Wi-Fi access points, switches, and routers across classrooms, computer labs, and libraries. It helps network administrators identify faulty devices and monitor network traffic.

Example: When students connect their laptops or smartphones to campus Wi-Fi, the network administrator can use SNMP monitoring software to check access point status and network performance.

3. Large Websites and Data Centers

Large websites and online services use SNMP to monitor network switches, routers, and other supported infrastructure devices in their data centers. It helps administrators detect high traffic, device failures, and network interface errors.

Example: When I use services such as Amazon or Flipkart, their network teams can use SNMP to monitor network infrastructure and help maintain reliable connectivity.

Conclusion: SNMP (Simple Network Management Protocol) helps network administrators monitor devices, collect performance information, detect faults, and maintain reliable network services. It monitors network infrastructure; it does not directly monitor the content or internal operation of apps such as Instagram or YouTube.

Task 5:

Create a table comparing FTP and HTTP across at least 5 points: port number, security, data transfer method, use cases, and authentication.

Answer:

No.	Comparison Point	FTP (File Transfer Protocol)	HTTP (Hypertext Transfer Protocol)
1	Port Number	Port 21 for control; port 20 is traditionally used for active-mode data transfer.	Port 80 by default.
2	Security	Standard FTP does not encrypt data or credentials. FTPS adds TLS encryption.	Standard HTTP does not encrypt data. HTTPS provides encryption using TLS.
3	Data Transfer Method	Transfers files between a client and a server.	Transfers web pages and other resources between a client and a web server.
4	Use Cases	Uploading, downloading, and managing files on a server.	Browsing websites, loading web pages, and communicating with web APIs.
5	Authentication	Can use a username and password; anonymous access may also be allowed.	Can use passwords, tokens, cookies, or other authentication methods; public pages may require no login.
6	Connection	Traditionally uses a separate control connection and data connections.	Uses request-response communication between client and server.
7	Common Software	FileZilla Client and FileZilla Server.	Web browsers such as Chrome, Edge, and Firefox.
8	Encrypted Version	FTPS or SFTP (a different protocol that uses SSH).	HTTPS.

Conclusion: FTP is mainly used for transferring and managing files between computers, whereas HTTP is mainly used to access websites and web services. For secure communication, FTPS or SFTP can be used for file transfers, and HTTPS is used for web traffic.